

Computer Science & IT

Database Management System



Relational Model & Normal Forms

Lecture No. 13



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Recap of Previous Lecture



Topic

Redundancy in relation because of FD

Topic

Second normal form (2NF)

Case ① $P.S.C.K \rightarrow N.P.A$
Not allowed

Topic

Third normal form (3NF)

Case ①, Case ② & Case ③
not allowed

In $X \rightarrow Y$
either X is a S.K
or
 Y is a prime attribute

Topic

Boyce codd normal form (BCNF)

All 5 cases are not
allowed

In $X \rightarrow Y$,
 X must be S.K

Topic

Decomposition of relation

Topics to be Covered



✓
Topic

Decomposition of relation



→ Possible types of non-trivial functional dependencies that may cause redundancy in the relation

X → Y

- | | | | |
|---------|----------------------------------|---|------------------------------|
| Case ①: | (P.S.C.K) Proper Subset of a C.K | → | Non Prime Attribute (N.P.A.) |
| Case ②: | (N.P.A.) Non prime Attribute | → | Non prime Attribute (N.P.A.) |
| Case ③: | (N.P.A + P.S.C.K.) | → | N.P.A. |
| Case ④: | P.S. of one C.K | → | P.S. of another C.K |
| Case ⑤: | (N.P.A + P.S. of one C.K) | → | P.S. of another C.K. |

Case Normal Form	Case① FD	Case② FD	Case③ FD	Case④ FD	Case⑤ FD
1NF	✓	✓	✓	✓	✓
2NF	✗ Not allowed	✓	✓	✓	✓
3NF	✗ Not allowed	✗ Not allowed	✗ Not allowed	✓	✓
BCNF	✗	✗	✗	✗	✗

} All types of FDs are allowed in 1NF

} FDs with Case①, Case② & Case③ are not allowed in 3NF

FDs of type Case① are not allowed in 2NF

All 5 types of FDs are not allowed in BCNF

#e.g. Given $R(ABCDE)$ and $F=\{AB \rightarrow C, C \rightarrow D, B \rightarrow E\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

CK: AB

P.A: $\{A, B\}$

N.P.A: $\{C, D, E\}$

$\underbrace{AB}_{S.K} \rightarrow C \quad \{BCNF\}$

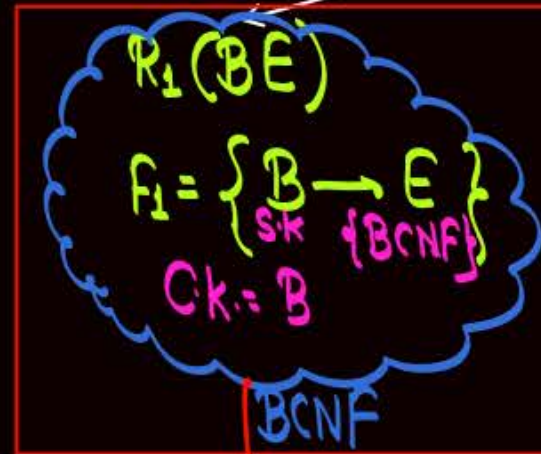
$\underbrace{C}_{N.P.A} \rightarrow \underbrace{D}_{N.P.A} \quad \{2NF\}$
Case 2

$\underbrace{B}_{P.S.C.K} \rightarrow \underbrace{E}_{N.P.A.} \quad \{1NF\}$
Case 1

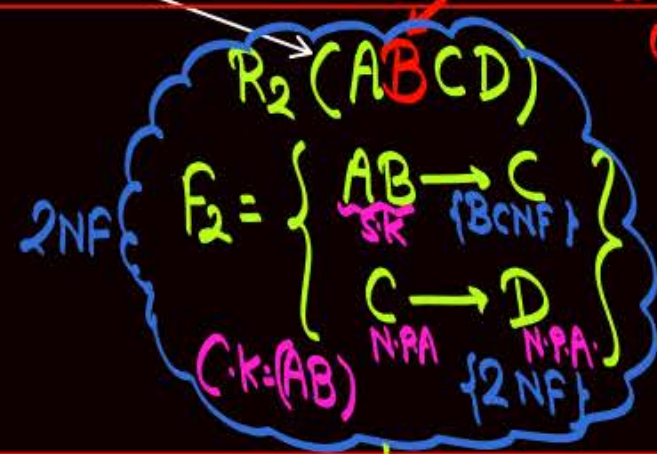
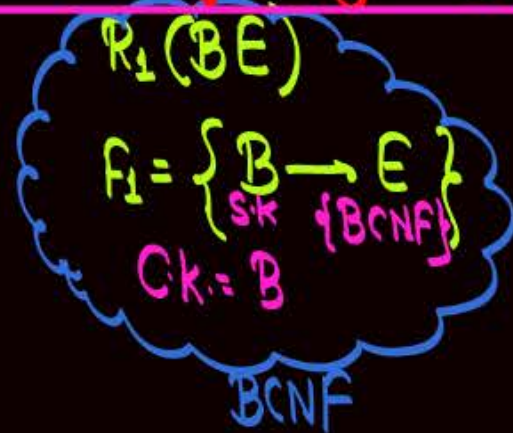
Normal form of Relation is "1NF"

$R(A, B, C, D, E)$ is not in 2NF
 because of FD " $B \rightarrow E$ "
 ∴ Decompose w.r.t. $B \rightarrow E$

$$(BE)^+ = \{B, E\}$$



already in BCNF



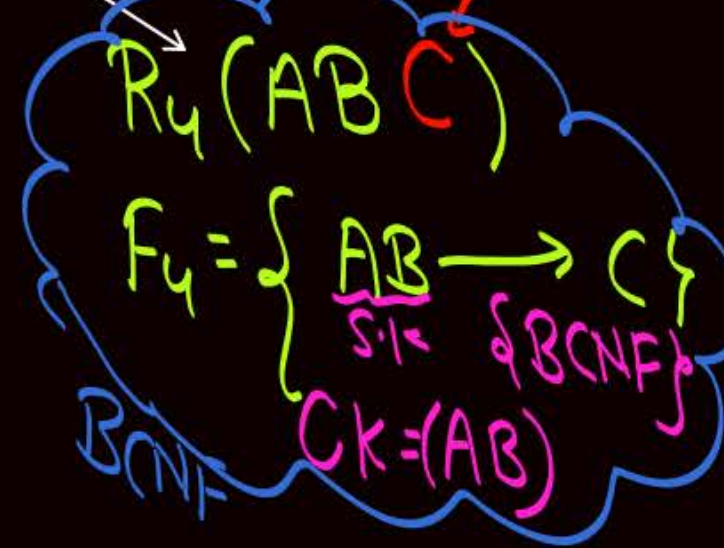
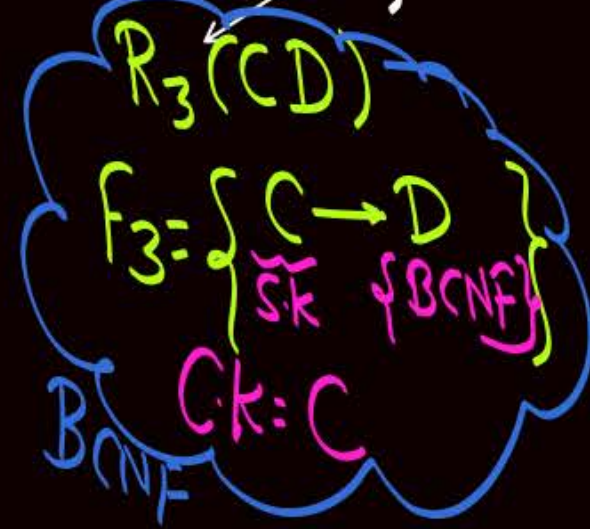
"B" is added
 to ensure
 lossless join
 decomposition

Overall
 normal form
 is = 2NF

+
 Lossless Join
 +
 Dep. preserving

$R_2(ABCD)$ is not in 3NF
 Because of $C \rightarrow D$.

∴ Decompose w.r.t. $C \rightarrow D$



added to ensure
 lossless join
 3NF as well as BCNF
 +
 Lossless join decomposition
 +
 Dep. preserving decomposition

H.W.

#e.g.



Given $R(ABCDEF)$ and $F = \{A \rightarrow BCDEF, BC \rightarrow ADEF, D \rightarrow E, E \rightarrow F\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

$$C.K. = A, (BC)$$

$$N.P.A = \{D, E, F\}$$

$$\underbrace{A}_{S.K.} \rightarrow BCDEF \quad \{BCNF\}$$

$$\underbrace{BC}_{S.K.} \rightarrow ADEF \quad \{BCNF\}$$

$$\underbrace{D}_{N.P.A} \rightarrow \underbrace{E}_{N.P.A} \quad \{2NF\}$$

$$\underbrace{E}_{N.P.A} \rightarrow \underbrace{F}_{N.P.A} \quad \{2NF\}$$

$R(ABCDEF)$ is not in 3NF because of
 $D \rightarrow E$ & $E \rightarrow F$
 let us decompose w.r.t $E \rightarrow F$

$(EF)^+ = \{E, F\}$

$R_1(EF)$

$F_1 = \{E \rightarrow F\}$
 ~~$\{BCNF\}$~~ BCNF
 $C_k = E$

$R_2(ABCDE)$

$F_2 = \{A \rightarrow BCDE, BC \rightarrow ADE, D \rightarrow E\}$
 ~~$\{BCNF\}$~~
 ~~$\{2NF\}$~~
 $C_k = A, (BC)$

for lossless Join

Overall normal form is 2NF
 + lossless Join
 + Dep preserving

$R_2(ABCDE)$ is not in 3NF
 because of $D \rightarrow E$
 Decompose w.r.t $D \rightarrow E$

$(DE)^+ = \{D, E\}$

$R_3(DE)$

$F_3 = \{D \rightarrow E\}$
 ~~$\{BCNF\}$~~
 $C_k = D$

$R_4(ABCD)$

$F_4 = \{A \rightarrow BCD, BC \rightarrow AD\}$
 ~~$\{BCNF\}$~~
 $C_k = A, (BC)$

for lossless

3NF + BCNF
 + lossless Join
 + Dep. preserving

In Union of $F_1 \cup F_3 \cup F_4$
 all FDK of F are preserved

H.W.
#e.g.



Given $R(ABCD)$ and $F=\{AB \rightarrow C, BC \rightarrow D\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

$\underbrace{AB}_{S.K} \rightarrow C$ {BCNF}

C.K = (AB)

P.A = {A, B}

N.P.A = {C, D}

$\begin{matrix} B & C & \rightarrow & D \\ \uparrow & \uparrow & & \uparrow \\ (P.S.C.K. + N.P.A) & & & (N.P.A) \end{matrix}$ {2NF}

Case ③ — Allowed in 2NF
but not allowed in 3NF

$R(ABCD)$ is not in 3NF
because of $BC \rightarrow D$

∴ Decompose w.r.t. $BC \rightarrow D$

$$(BCD)^+ = \{B, C, D\}$$

$R_1(BCD)$

$$F_1 = \left\{ \underbrace{BC}_{\text{S.K.}} \rightarrow D \right\}_{\text{BCNF}}$$

C.K.: (BC)

BCNF

$R_2(ABC)$

$$F_2 = \left\{ \underbrace{AB}_{\text{S.K.}} \rightarrow C \right\}_{\text{BCNF}}$$

C.K.: (AB)

BCNF

for lossless Join

3NF + BCNF

+
lossless Join

+
Dep. preserving

H.W.



#e.g.

Given $R(ABCDEFGH IJ)$ and $F = \{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

$CK = (AB)$

$AB \rightarrow C$
 $\underbrace{S.K.}_{\{BCNF\}}$

$A \rightarrow DE$
 $P.S.K. \rightarrow N.P.A. \text{ Case ①}$
 $\{1NF\}$

$D \rightarrow IJ$
 $N.P.A. \rightarrow N.P.A. \text{ Case ②}$
 $\{2NF\}$

$B \rightarrow F$
 $P.S.K. \rightarrow N.P.A. \text{ Case ①}$
 $\{1NF\}$

$F \rightarrow GH$
 $N.P.A. \rightarrow N.P.A. \text{ Case ②}$
 $\{2NF\}$

$R(AB \underline{CDE} FGHIJ)$

Let us first decompose w.r.t. $B \rightarrow F$

$(BF)^+ = \{B, F, G, H\}$

$R_1(BF)$

$F_1 = \left\{ \underset{\text{PK}}{B} \rightarrow F \text{ (BCNF)} \right\}$

C.K.: B

$R_2(\underline{AB} CDEFGHIJ)$

for lossless

$F_2 = \left\{ \begin{array}{l} AB \rightarrow C \\ A \rightarrow DE \\ D \rightarrow IJ \\ B \rightarrow GH \end{array} \right\}$

$(F)^+ \text{ w.r.t. } F_1 \cup F_2 = \{F\}$

ie $G, H \notin (F)^+ \text{ w.r.t. } F_1 \cup F_2$

Hence, $F \rightarrow GH$ is lost.

Because $(BF)^+ \text{ w.r.t. } F = \{B, F, G, H\}$
∴ Include G & H also in R_1

R(AB CDEFGHIJ)

Let us first decompose wrt $B \rightarrow F$

$(BF)^+ = \{B, F, G, H\}$

for lossless

1NF
+
lossless Join
+
Dep. Preserving

$R_1(B, F, G, H)$

$F_1 = \left\{ \begin{array}{l} B \rightarrow F \text{ (BCNF)} \\ F \rightarrow GH \text{ (2NF)} \end{array} \right\}$
 CK: B
 PA: {B}
 NPA: {F, G, H}

2NF

$R_2(AB CDE \cancel{G} \cancel{H} IJ)$

$F_2 = \left\{ \begin{array}{l} AB \rightarrow C \text{ (BCNF)} \\ A \rightarrow DE \text{ (1NF)} \\ D \rightarrow IJ \text{ (2NF)} \end{array} \right\}$
 CK: AB

1NF

$\rightarrow R_2$ is in 1NF because of $A \rightarrow DE$
 $\therefore$ Decompose wrt $A \rightarrow DE$

$(ADE)^+ = \{A, D, E, I, J\}$

$R_3(A, D, E, I, J)$

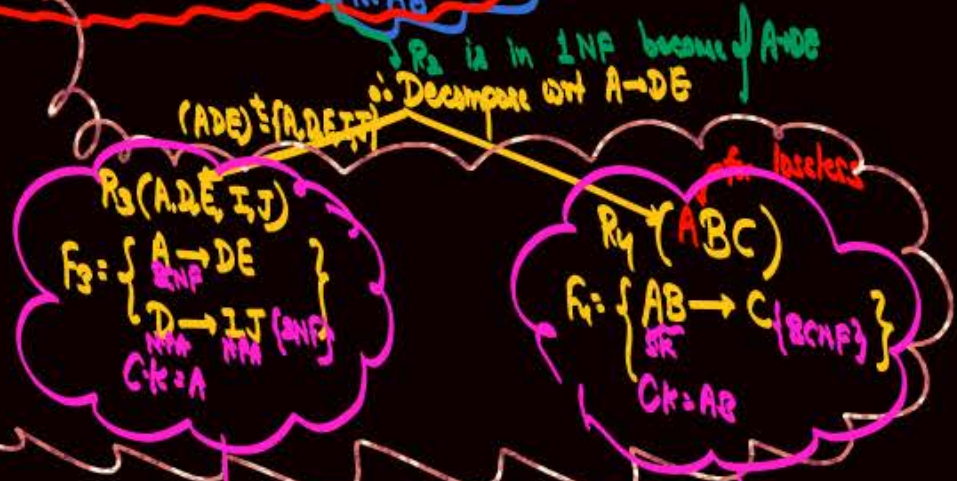
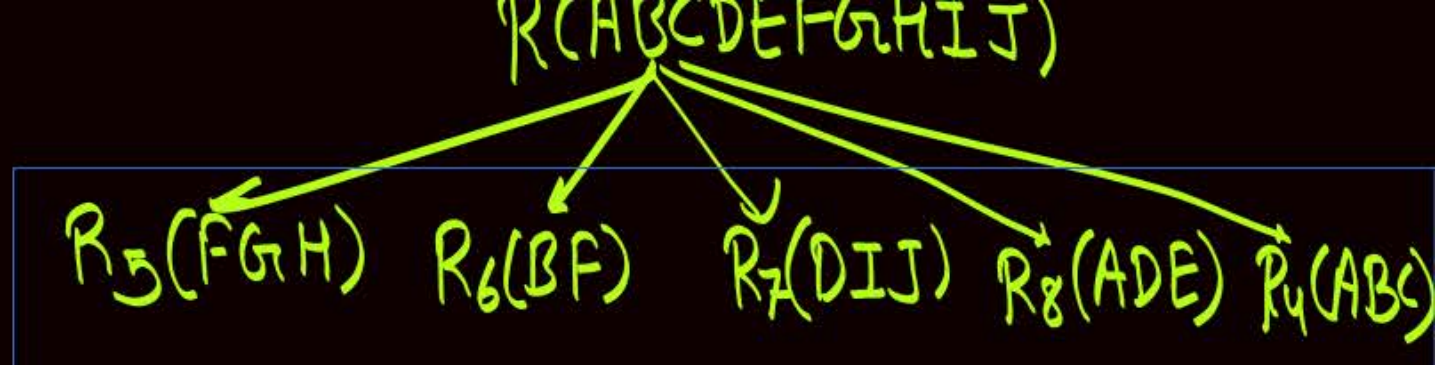
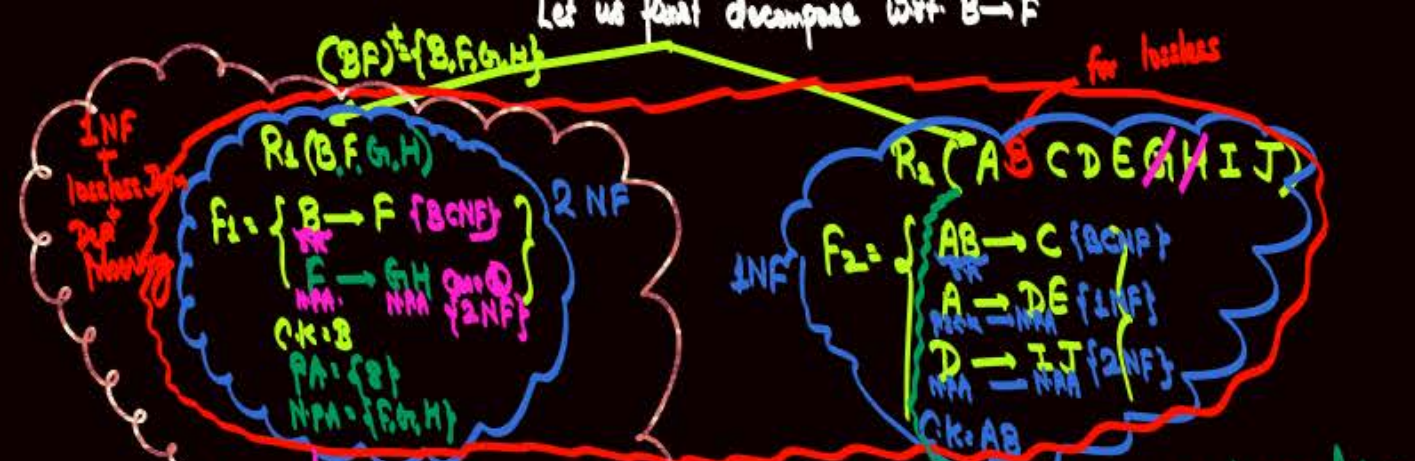
$F_3 = \left\{ \begin{array}{l} A \rightarrow DE \text{ (BCNF)} \\ D \rightarrow IJ \text{ (2NF)} \end{array} \right\}$
 CK: A

for lossless

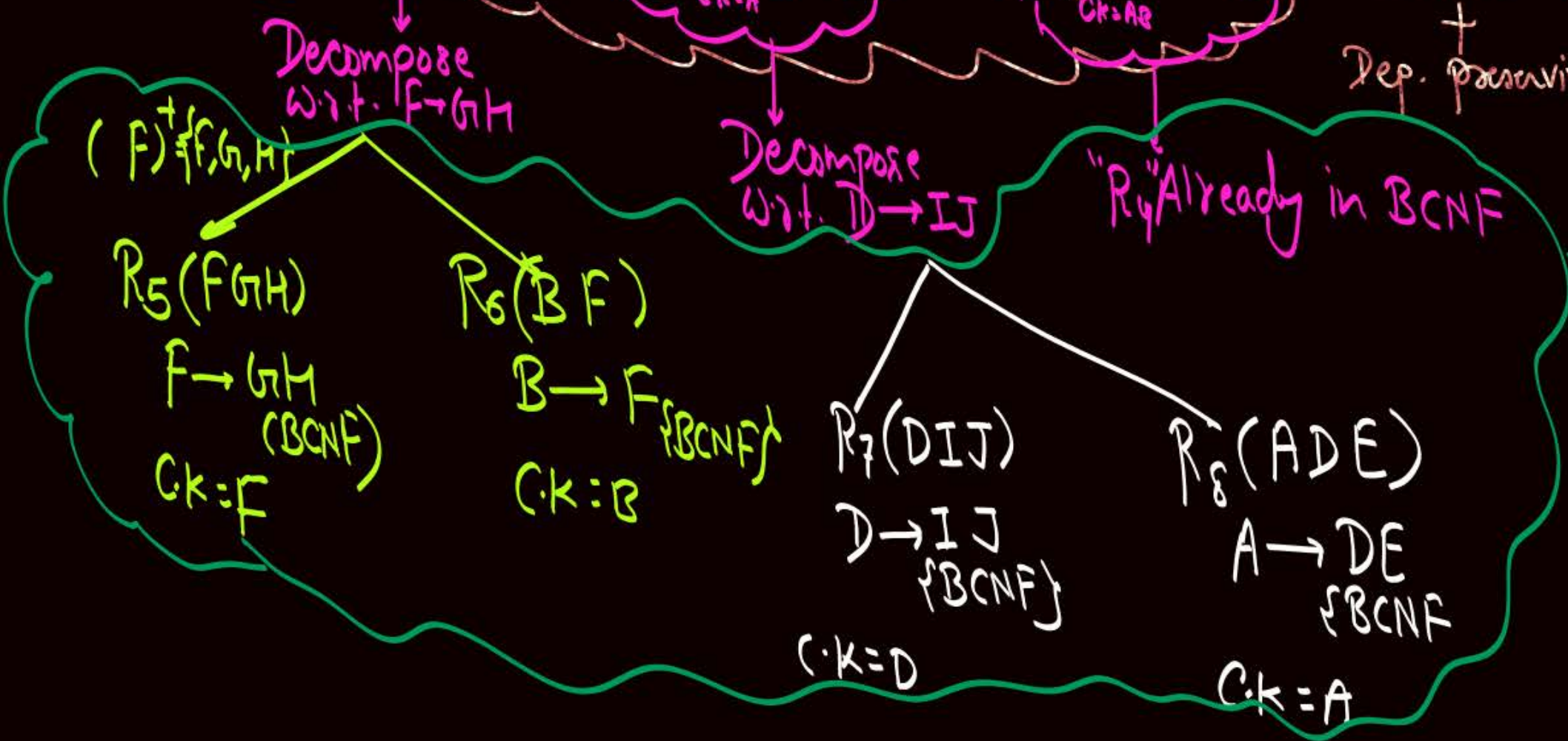
$R_4(ABC)$

$F_4 = \left\{ AB \rightarrow C \text{ (BCNF)} \right\}$
 CK: AB

overall normal form
 is = 2NF
 +
 lossless Join
 +
 Dep. preserving



overall normal form is = 2NF + lossless Join + Dep. preserving



3NF + BCNF + lossless Join + Dep. preserving

H.W.



#e.g.

Given $R(ABDLPT)$ and $F=\{B \rightarrow PT, T \rightarrow L, A \rightarrow D\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

Try by yourself, it is exactly same as
Previous relation

H.W.

C.K. = (AB)



#e.g. Given $R(ABDLPT)$ and $F = \{B \rightarrow PT, T \rightarrow L, A \rightarrow D\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.



Note:-

If FD set of a relation is Empty, i.e. No non-trivial FD exists in that relation, then that relation will always be in BCNF.

{ No non-trivial FD exist in the relation,
∴ no redundancy can exist in the relation }
because of FD and hence normal form
of relation will be in BCNF

Note:- If there exists no non-trivial FD in the relation, then Candidate key of that relation will be formed by combining all the attributes of that relation.

H.W.



#e.g.

Given $R(ABCDE)$ and $F:\{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

C.K = A, E, CD, BC

$\underset{\text{S.K}}{A} \rightarrow BC \quad \{BCNF\}$

$\underset{\text{S.K}}{CD} \rightarrow E \quad \{BCNF\}$

$\underset{\text{S.K}}{E} \rightarrow A \quad \{BCNF\}$

$B \xrightarrow{\text{Case 4}} D \quad \{3NF\}$
P.S. of one C.K. Proper subset of another C.K.

$R(ABCDE)$ is not in BCNF becoz of $B \rightarrow D$
∴ Decompose w.r.t $B \rightarrow D$

$$(BD)^+ = \{BD\}$$

$R_1(BD)$

$$F_1 = \{ B \rightarrow D \}$$

PK: B

BCNF

BCNF

$R_2(ABCE)$

$$F_2 = \begin{cases} A \rightarrow BCE & \{BCNF\} \\ E \rightarrow ABC & \{BCNF\} \\ BC \rightarrow AE & \{BCNF\} \end{cases}$$

PK: A, (BC), E

for lossless

BCNF

To preserve $CD \rightarrow E$
We can create a
new subrelation

$R_3(CDE)$

$$F_3 = \{ CD \rightarrow E \}$$

PK: CD

$$(CD)^+ \text{ w.r.t } F_1 \cup F_2 = \{CD\}$$

∴ $CD \rightarrow E$ will be
lost.

BCNF
+
Lossless Join
+
Dep. preserving



Given $R(ABCD)$ and $F=\{AB \rightarrow CD, D \rightarrow A\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

Q: Are we able to decompose the relation into BCNF while all the dependencies are preserved?
=

THANK - YOU